ASHISH VISHWAKARMA

Application & Product Security · AI/ML Security · VAPT · Mobile & Malware Reverse Engineering
Bangalore, India

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Profile

Security engineer with 4+ years across application & product security, mobile and malware reverse engineering, and AI/ML security. Currently reverse engineering Android malware campaigns at ecosystem scale; previously led end-to-end web, API and mobile VAPT plus GenAI red-teaming for Tier-1 banking and fintech clients. Comfortable from Burp and secure code review down to Smali/DEX and native ARM binaries.

Experience

Google Operations Center, Bangalore, IN — July 2025–Present
Security Engineer Specialist

- Reverse engineer ML-clustered Android malware campaigns at ecosystem scale, surfacing policy-enforcement gaps for Potentially Harmful Applications (PHAs) across a high number of user installs.
- Dissect Smali/DEX bytecode and native JNI payloads (`.so`/ELF, ARM) in IDA Pro, Ghidra and radare2; defeat commercial packers, string encryption and anti-analysis checks.
- Analyse obfuscated APKs and cross-platform binaries (Unity, Cocos2d-x, Flutter), separating malicious payloads from engine artifacts to cut false-positive triage volume.
- Map campaign infrastructure through encrypted HTTP redirection chains, server-side cloaking and weaponised C2 endpoints used to evade automated scanners.
- Ship production SQL heuristic detection signatures over structural and network IoCs, automating at-scale identification of many evolving malware families.

Appknox, Bangalore, IN — August 2024–July 2025
Security Analyst

- Led end-to-end mobile and web VAPT (Android, iOS, REST/GraphQL) for Tier-1 banking and fintech clients against OWASP MASVS and ASVS.
- Reverse engineered and instrumented hardened Android/iOS applications with Frida, Objection and Jadx, bypassing root/jailbreak detection, SSL pinning and anti-tamper controls.
- Audited AWS environments, remediating IAM privilege-escalation paths, unauthenticated S3 policies and hardcoded secrets across microservices.
- Red-teamed production GenAI and RAG applications against the OWASP LLM Top 10, proving unauthorised internal data retrieval via indirect prompt injection.
- Built Python tooling with ML classifiers (NLP, Random Forest) to triage SAST/DAST output, reducing assessment turnaround by 30%.

EC-Council, Hyderabad, IN — December 2023–August 2024
Security Researcher

- Architected vulnerable-by-design AWS and Azure environments simulating cloud Active Directory attack and privilege-escalation chains for enterprise certification training.
- Authored CTF challenges and vulnerable microservices covering cryptographic flaws, token manipulation and binary analysis; automated lab deployment with Docker.
- Designed hands-on curriculum for CEH and CHFI covering modern web exploitation and API abuse.
- Evaluated emerging offensive tooling against enterprise EDR to model APT attack lifecycles.

HackerHeal, Bangalore, IN — November 2022–December 2023
Cybersecurity Analyst

- Conducted digital forensics and OSINT investigations supporting cybercrime response; tracked threat-actor infrastructure, domains and C2 servers using Maltego and Shodan.
- Executed full-scope web and network penetration tests (Burp Suite Pro, Metasploit, Nmap), identifying business-logic flaws and authentication bypasses.
- Extracted IoCs from packet captures (Wireshark) and authentication logs; authored forensic reports on active phishing syndicates.

Intel Corporation, Remote, IN — May 2022–July 2022
Computer Vision Security & ML Intern

- Built real-time edge computer-vision inference pipelines (YOLO) for multi-object detection and tracking, optimising stream ingestion latency under edge compute constraints.

Technical Skills

Application & API Security: Web, REST & GraphQL VAPT, Burp Suite Pro, OWASP ASVS, secure code review (Python, Go, JavaScript), SAST/DAST triage, threat modelling.
Mobile Security: Android & iOS assessment, OWASP MASVS, Frida, Objection, Jadx, root/jailbreak & SSL-pinning bypass, anti-tamper defeat.
AI/ML Security: OWASP Top 10 for LLMs, prompt injection, RAG & agent red-teaming, adversarial robustness, ML-assisted triage (NLP, Random Forest).
Reverse Engineering: IDA Pro, Ghidra, radare2, Smali/DEX, native JNI (`.so`/ELF), ARM & x86 assembly, unpacking, deobfuscation, anti-analysis defeat.
Cloud & Infrastructure: AWS (IAM, S3, KMS), GCP, Azure AD, Docker, Kubernetes, Terraform IaC.
Detection & Automation: Heuristic signature engineering, IoC extraction, malware clustering & triage pipelines, SQL, Python tooling.
Core & Compliance: Python, Go, C/C++, x86/ARM assembly; OWASP ASVS/MASVS, ISO 27001, DORA/GDPR. English (Professional), Hindi (Native).

Certifications

Offensive & Mobile: Burp Suite Certified Practitioner (BSCP), eMAPT, eWPT, Certified API Penetration Tester.
Cloud & AI: AWS Certified Security – Specialty (SCS-C02), AI/ML Pentesting Certified (The SecOps Group).
Additional: CEH v12, SC-900, ISC2 CC, ISO 27001 InfoSec Associate.

Education

Parul University, Gujarat, Vadodara, IN — August 2020–April 2024
Bachelor of Technology in Computer Engineering (Cybersecurity)

Honors & Recognition

- **Security Hall of Fame – Harvard University & NASA:** Acknowledged by both organisations' vulnerability disclosure programs for responsibly reporting security flaws in public-facing infrastructure.
- **Author:** *Hacker's Havoc: A Blueprint for Dominance* (Amazon).